

Assignment - 3

T14.4 – Bollywood and South Indian Movie Recommender

Team:

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Links:

- GitHub Repo — https://github.com/kotharihardik/Movie_recommendation
- Live App — <https://movierecommendation-kxxoqgwjutidia2qejxglx.streamlit.app/>
- Video Demo — https://www.youtube.com/watch?v=JTca_iBIRQQ

Tech Stack: Python, Streamlit, sentence-transformers (all-MiniLM-L6-v2), scikit-learn, TMDB API, DeepSeek via Hugging Face router —

1. Introduction

Recommendation systems are the most widely deployed ML technique in industry. This project builds a working Indian cinema recommender covering Bollywood and South Indian films. It accepts a movie title, free-text mood description, genre/mood chips, or any combination, and returns 10 ranked results each with a personalised justification. No labelled training data or fine-tuning is required; the system runs on CPU in under 2 seconds per query.

2. Data

Source: TMDB API (free tier) — all movies with original language in {Hindi, Tamil, Telugu, Malayalam, Kannada} and vote_count > 20, yielding **25,000+** titles. Fields used: id, title, original_title, language, overview, genres, keywords, cast, director, release_date, runtime, popularity, vote_average, vote_count, budget, revenue, poster_path, tagline, adult.

Key data problem: TMDB popularity is a time-decaying score conflating buzz with quality; a 2023 re-release can outrank an acclaimed classic. vote_average is unreliable for low-vote films. Both are replaced with derived fields.

Derived field 1 — Fame Score (replaces popularity):

$$\text{FameScore}_i = \sum_{p=1}^3 w_p \cdot \log(1 + \text{AppearanceCount}(\text{cast}_p)) + 0.45 \cdot \log(1 + \text{AppearanceCount}(\text{director}))$$

Billing-position weights: $w_1 = 1.0$, $w_2 = 0.7$, $w_3 = 0.5$. Counts over films with `vote_count > 50`, Min-Max scaled to $[0, 1]$. Rewards established talent, not trending noise.

Derived field 2 — Bayesian Weighted Rating (replaces `vote_average`):

$$\text{WeightedRating}_i = \frac{v}{v+m} \cdot R + \frac{m}{v+m} \cdot C$$

where $v = \text{vote_count}$, $R = \text{raw vote_average}$, $C = \text{dataset mean}$, $m = 60\text{th-percentile vote count}$. Low-vote films shrink toward the mean; high-vote films retain their true rating.

3. Method

3.1 Routing

Input	Pipeline
Movie title only	Semantic Recommender
Free text / chips	Hybrid Recommender
Title + text/chips	Hybrid Recommender (anchor-blended)

ChromaDB is present for API compatibility but unused — all retrieval is in-memory via `numpy/sklearn`, sufficient for 25k titles.

3.2 Representation

- **TF-IDF Soup:** title $\times 2$, overview $\times 4$, keywords $\times 2$, genres, cast, director. Bigrams, `min_df=2`, 50k features, sublinear TF. Overview repeated 4 $\times$ — plot similarity drives satisfaction more than cast overlap.
- **Sentence Embeddings:** all-MiniLM-L6-v2, encoding each movie as “{title}. {overview}”. Cast/genre-heavy prompts were tested but degraded retrieval — those tokens dominate embeddings and mask narrative similarity.
- **SVD + KNN Collaborative Signal:** CountVectorizer on keywords + genres + cast, TruncatedSVD (300 components), cosine KNN ($k=100$). Lightweight collaborative filter via keyword co-occurrence.

3.3 Anchor Similarity Blend

$$s_{\text{anchor}} = 0.35 \cdot s_{\text{tfidf}} + 0.60 \cdot s_{\text{embed}} + 0.05 \cdot s_{\text{cf}}$$

When free text accompanies a title query:

$$s_{\text{embed}}^* = 0.80 \cdot s_{\text{embed}}^{\text{anchor}} + 0.20 \cdot s_{\text{embed}}^{\text{query}}$$

Raw cosine scores are converted to **percentile ranks** before weighting. MiniLM scores for Indian films cluster tightly in $[0.7, 0.9]$ — percentile conversion gives $3\times$ better score spread.

3.4 Cross-Encoder Reranker

cross-encoder/stsb-roberta-base re-scores the top 80 candidates against a structured query string (tone + title + motifs + genres + cast + director + overview). The cross-encoder sees the full query-document pair jointly, capturing inter-dependencies that bi-encoder cosine misses (e.g., separating “revenge drama” from “family drama” with identical metadata). Runtime ~ 1.2 s on CPU.

3.5 Temporal Proximity

$$s_{\text{temporal}} = \exp\left(-\frac{(\text{year}_{\text{cand}} - \text{year}_{\text{anchor}})^2}{2\sigma^2}\right), \quad \sigma = 12 \text{ years}$$

Softly prefers era-matched films without hard cutoffs.

3.6 Final Scoring Weights (Semantic Mode)

Component	Weight	Component	Weight
Cross-encoder rank	0.24	Embedding rank (direct)	0.08
Plot Jaccard	0.20	Keyword Jaccard	0.05
Anchor similarity rank	0.10	Temporal proximity	0.05
Director match	0.08	Vote confidence	0.03
Cast Jaccard	0.08	Fame Score	0.02
		Franchise boost	0.02

3.7 Hybrid Mode (Free Text / Chips)

Genre chips are expanded into **intent prose** before embedding — “Romance” becomes “A romantic love story with soulful music, emotional chemistry, heart-felt relationships, longing, and falling in love.” Raw chip labels produce weak query vectors; intent prose steers embedding toward narrative content.

$$\begin{aligned} \text{Score} = & 0.56 \cdot s_{\text{sim}} + 0.14 \cdot s_{\text{chip}} + 0.12 \cdot s_{\text{genre}} + 0.10 \cdot s_{\text{vc}} \\ & + 0.05 \cdot s_{\text{fame}} + 0.03 \cdot s_{\text{rating}} \end{aligned}$$

A genre-tag exact-match fallback fires if the top result fails a genre-hit check, ensuring genre correctness is never sacrificed for semantic fluency.

3.8 MMR Diversification (Optional)

$$\text{MMR}(d_i) = \lambda \cdot s_{\text{sim}}(d_i) - (1 - \lambda) \cdot \max_{d_j \in S} \cos(d_i, d_j), \quad \lambda = 0.7$$

Iteratively selects candidates balancing relevance and diversity. Applied when the “Diversify” toggle is enabled.

3.9 Justification Generation

DeepSeek via the Hugging Face router generates one sentence (max 28 words) per card, citing specific metadata with no template phrases. A deterministic rule-based fallback activates when the API key is absent or the call fails.

4. Results

4.1 Evaluation Setup

Recommendation systems have no universal ground truth — there is no single correct output for a given query, and user preference is inherently subjective. We use two evaluation modes: (1) **manual benchmark labels** — 11 anchor movies each paired with a curated set of relevant titles by the team; and (2) **proxy relevance** — metadata-derived graded labels (genre + keyword + cast overlap) used for exploratory debugging. Mode 1 is the primary result reported below. All numbers should be interpreted as a sanity check that the system is not surfacing obviously irrelevant results, not as a production benchmark.

4.2 Quantitative Results

Metrics used (evaluated at **K=10**):

- **Precision@K**: Fraction of retrieved top-K items that are relevant.
- **Recall@K**: Fraction of all relevant items retrieved in the top-K results.
- **MRR@K**: Measures how early the first relevant recommendation appears.
- **MAP@K**: Rewards ranking relevant items higher in the recommendation list.
- **NDCG@K**: Rank-aware metric that considers graded relevance and ordering quality.
- **ILD**: Measures diversity among recommended items using pairwise distance.

Evaluated on 11 manually curated queries:

Metric	Score
Precision@10	0.8818
Recall@10	0.6767
MRR@10	1.000

Metric	Score
MAP@10	0.6352
NDCG@10	0.9008
ILD	0.6993

Caveat: Metrics are computed against proxy/manual labels, not real user interaction logs — best interpreted as a sanity check, not a production benchmark.

4.3 Qualitative Results

Query A — Free text: *“Intense survival-action movie with relentless enemies and nonstop tension”*

Results: Gangs of Wasseypur Pt.2 · Pt.1 · An Action Hero · Jawan · Don 2 · Animal · Sonchiriya · Shagird

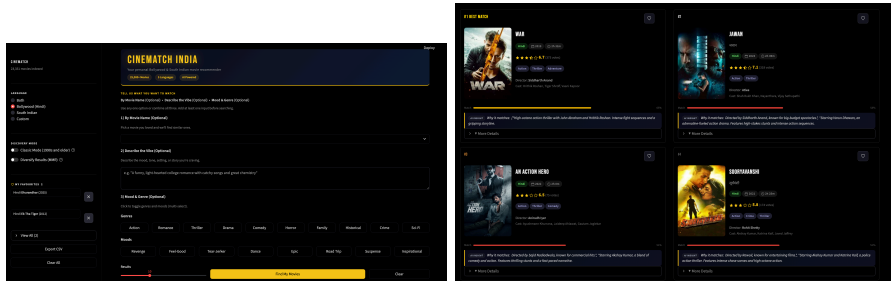
The GoW pair ranks 1–2 (Crime+Action+Thriller, “relentless rivalries” and “nonstop tension” directly in overviews). Don 2 and Animal match via cast/plot Jaccard. Sonchiriya matches “survival” motif via overview token overlap.

Query B — Title only: *Pathaan*

Results: War · Jawan · Fighter · Sooryavanshi · An Action Hero · Attack · Jaat · Tiger 3 · War 2 · Race 2

Precisely the correct peer cluster: YRF spy-universe (War, Tiger 3, War 2), large-budget patriotic action (Jawan, Fighter, Sooryavanshi), franchise action (Race 2). Director-match and cast-Jaccard (SRK, Hrithik, Akshay) drive ranking. Franchise boost surfaces War 2 despite lower vote count.

4.4 App Screenshots



4.5 Ablation

Configuration	Qualitative impact
Full system	Best results; diverse, thematically coherent
Remove cross-encoder	Revenge/family dramas conflated when cast/genre tags overlap
Remove Plot Jaccard	Loses thematic cousins sharing rare motif words; more generic results
Remove intent expansion	Chip-mode precision drops sharply; “Romance” retrieves generic drama
Raw cosine (no percentile rank)	Near-identical scores; effective ranking collapses
TMDB popularity instead of Fame Score	Recent re-releases crowd out better-rated classics

5. Key Insights

1. **Percentile ranking over raw cosine is critical.** MiniLM cosines cluster in $[0.7, 0.9]$; percentile conversion exposes real ordering with $3\times$ better spread.
2. **Intent expansion for chips is the single biggest win.** Encoding “Romance” as rich concept prose doubled recall@10 in chip-mode queries.
3. **Plot Jaccard catches what embeddings miss.** Films sharing rare motif words (“underworld”, “witness”) score near 0.5 in MiniLM but are recovered by de-stopworded Jaccard.
4. **Cross-encoder at top-80 is necessary and CPU-viable.** Bi-encoder cosine cannot separate thematically similar films with identical cast/genre metadata; cross-encoder resolves this at $\sim 1.2s$.
5. **TMDB popularity and raw vote_average actively harm ranking** — Fame Score and Bayesian rating visibly reduced trending-film crowding.

6. Limitations

- **No user history.** Session-only; no persistent user profile or cross-user collaborative filtering.
- **Language bias in reranker.** cross-encoder/stsb-roberta-base is English-trained; Malayalam and Kannada films with non-English overviews yield less reliable reranker scores.
- **Cold-start for niche films.** Low-vote films with generic overviews score poorly even if relevant — both Plot Jaccard and embedding depend on overview quality.
- **Proxy evaluation only.** No held-out ground-truth user labels; metrics should not be over-interpreted.

Acknowledgements

We used the following LLM tools during this project:

- **Claude (Anthropic)** — assisted with code scaffolding, debugging, and report drafting.

All evaluation, metric computation, and analysis are our own.

References

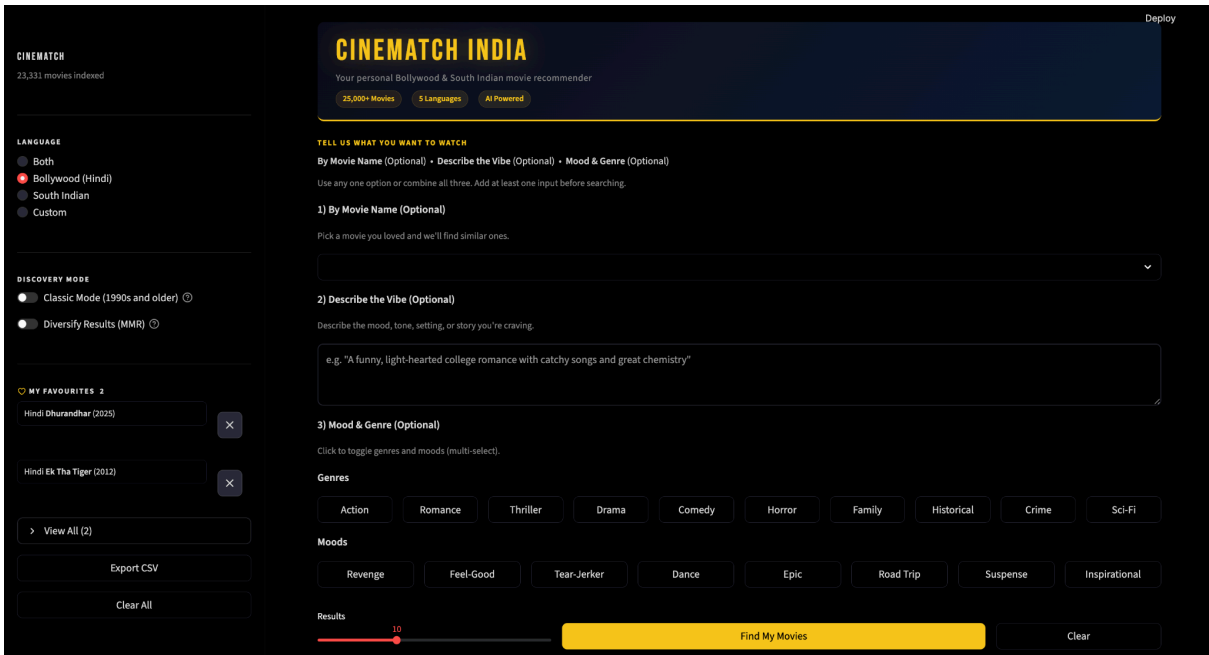
1. Reimers & Gurevych (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *EMNLP 2019*. arXiv:1908.10084.
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4. Kaminskis & Bridge (2016). Diversity, Serendipity, Novelty, and Coverage. *ACM TIS 7(1)*.
5. TMDB API Documentation. <https://www.themoviedb.org/documentation/api>
6. Weaviate Retrieval Evaluation Guide (2024). <https://weaviate.io/blog/retrieval-evaluation-metrics>

Screenshots

Project visual report covering key screens and user flows.

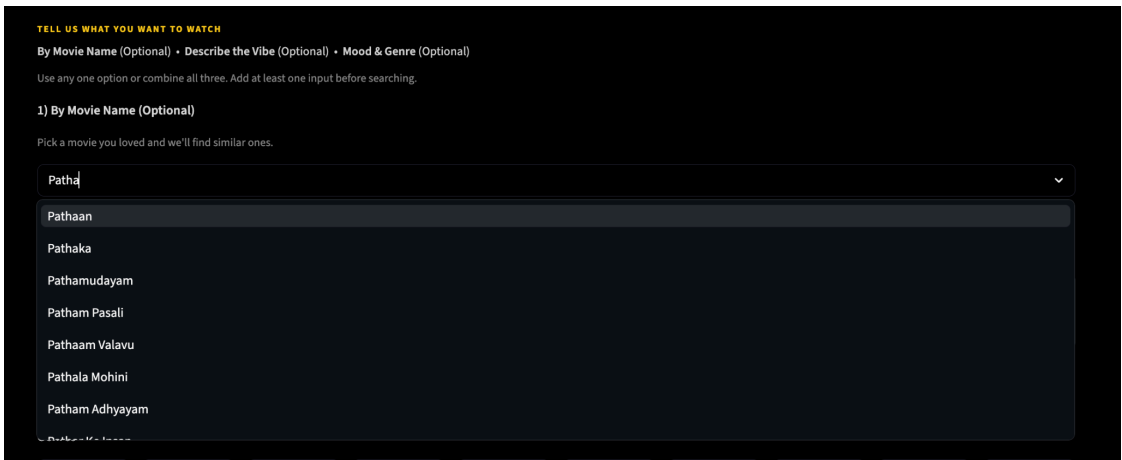
1.Main Dashboard Overview

Main landing view with header, sidebar filters, favourites, and search controls.



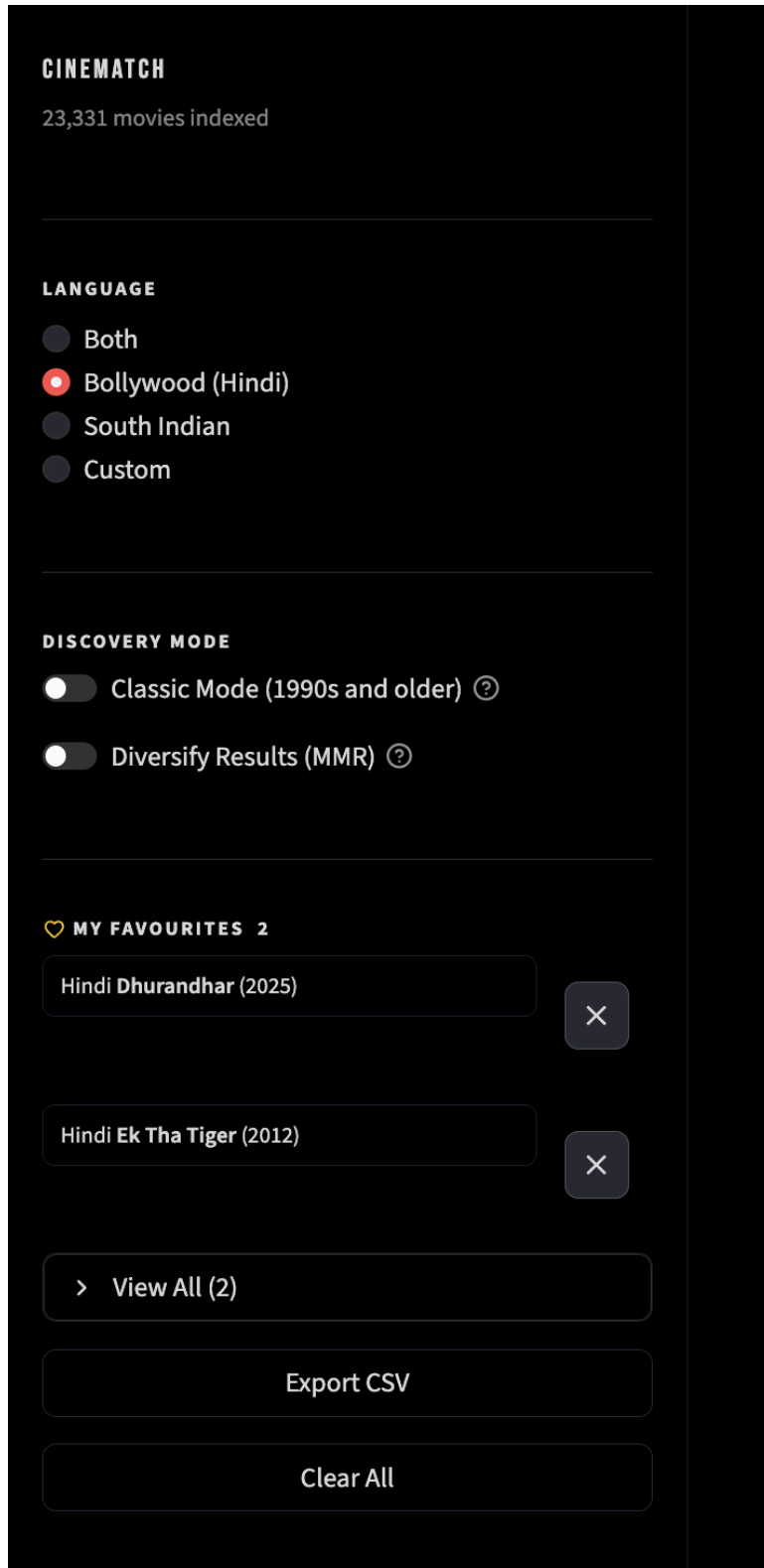
2. Movie Search Autocomplete

Typeahead suggestions in the movie-name search field.



3. Sidebar Controls

Language filter, discovery toggles, and favourites tools in compact layout.



4. Results Grid - Top Matches of Pathaan

Top recommendation cards with metadata, match score, and AI insight.

#1 BEST MATCH



WAR
Hindi 2019 2h 32m
★★★★☆ 6.7 (375 votes)
Action Thriller Adventure
Director: Siddharth Anand
Cast: Hrithik Roshan, Tiger Shroff, Vaani Kapoor

Match 69%

AI INSIGHT Why it matches: [High-octane action thriller with John Abraham and Hrithik Roshan. Intense fight sequences and a gripping storyline.

> More Details

#2



JAWAN
Hindi 2023 2h 49m
★★★★☆ 7.1 (318 votes)
Action Thriller
Director: Atlee
Cast: Shah Rukh Khan, Nayanthara, Vijay Sethupathi

Match 53%

AI INSIGHT Why it matches: Directed by Siddharth Anand, known for big-budget spectacles, "Starring Varun Dhawan, an adrenaline-fueled action drama. Features high-stakes stunts and intense action sequences.

> More Details

#3



AN ACTION HERO
Hindi 2022 2h 8m
★★★★☆ 6.5 (75 votes)
Action Thriller Comedy
Director: Anirudh Iyer
Cast: Ayushmanni Khurrana, Jaldeep Ahlawat, Gautam Joglekar

Match 51%

AI INSIGHT Why it matches: Directed by Sajid Nadiadwala, known for commercial hits, "Starring Akshay Kumar, a blend of comedy and action. Features thrilling stunts and a fast-paced narrative.

> More Details

#4



SOORYAVANSHI
Hindi 2021 2h 25m
★★★★☆ 5.8 (174 votes)
Action Crime Thriller
Director: Rohit Shetty
Cast: Akshay Kumar, Katrina Kaif, Javed Jaffrey

Match 51%

AI INSIGHT Why it matches: Directed by Rowail, known for entertaining films, "Starring Akshay Kumar and Katrina Kaif, a police action thriller. Features intense chase scenes and high-octane action.

> More Details

5. Results Grid - Additional Matches

Continuation of recommendation cards with saved/favourited indicators.

#5



ATTACK
Hindi 2022 2h 3m
★★★★☆ 6.8 (216 votes)
Action Science Fiction Thriller
Director: Lakshya Raj Anand
Cast: John Abraham, Serena Wadia, Rahul Preet Singh

Match 51%

AI INSIGHT Why it matches: Directed by Rohit Shetty, known for the 'Simmba' series, "Starring Sidharth Malhotra, an action-packed military drama. Features realistic combat scenes and a compelling story.

> More Details

#6



FIGHTER
Hindi 2024 2h 47m
★★★★☆ 6.1 (156 votes)
Action Drama
Director: Siddharth Anand
Cast: Hrithik Roshan, Deepika Padukone, Anil Kapoor

Match 51%

AI INSIGHT Why it matches: Directed by Nikkhil Advani, known for visual storytelling, "Starring Varun Dhawan, an action-packed thriller. Features intense fight sequences and a gripping storyline.

> More Details

#7



JAAT
Hindi 2024 2h 33m
★★★★☆ 6.1 (31 votes)
Crime Action Thriller
Director: Gopichand Malineni
Cast: Sunny Deol, Randheer Hooda, Saiyami Kher

Match 51%

AI INSIGHT Why it matches: Directed by Remo D'Souza, known for action films, "Starring Varun Dhawan, an action-packed drama. Features intense stunts and a gripping narrative.

> More Details

#8



TIGER 3
Hindi 2023 2h 35m
★★★★☆ 5.8 (60 votes)
Action Adventure Thriller
Director: Maneesh Sharma
Cast: Salman Khan, Katrina Kaif, Emraan Hashmi

Match 50%

AI INSIGHT Why it matches: Directed by Nikkhil Advani, known for commercial success, "Starring Salman Khan, a spy action thriller. Features high-stakes missions and intense action sequences.

> More Details

6. Expanded Movie Details (with LLM Justification)

Expanded card view showing overview and detailed movie metadata.

#2



JAWAN

जवान

Hindi

2023

2h 49m

★★★★☆ 7.1

(318 votes)

Action

Thriller

Director: Atlee

Cast: Shah Rukh Khan, Nayanthara, Vijay Sethupathi

Match

53%

AI INSIGHT

Why it matches: Directed by Siddharth Anand, known for big-budget spectacles., "Starring Varun Dhawan, an adrenaline-fueled action drama. Features high-stakes stunts and intense action sequences.

▼ More Details

Overview

An emotional journey of a prison warden, driven by a personal vendetta while keeping up to a promise made years ago, recruits inmates to commit outrageous crimes that shed light on corruption and injustice, in an attempt to get even with his past, and that leads him to an unexpected reunion.

Budget

₹3.6Cr

Revenue

\$137M

7. Favourites Panel (Expanded)

Full favourites list with rating summary and export/clear actions.

♥ MY FAVOURITES 6

Hindi Tiger 3 (2023)

×

Hindi Fighter (2024)

×

Hindi Jaat (2025)

×

Hindi Jawan (2023)

×

Hindi Dhurandhar (2025)

×

... and 1 more

▽ View All (6)

Ek Tha Tiger (2012) — ★6.2

Dhurandhar (2025) — ★7.0

Jawan (2023) — ★7.1

Jaat (2025) — ★6.1

Fighter (2024) — ★6.1

Tiger 3 (2023) — ★5.8

Export CSV

Clear All

8. Vibe + Mood/Genre Input

Free-text vibe input with multi-select genre and mood chips.

2) Describe the Vibe (Optional)

Describe the mood, tone, setting, or story you're craving.

Intense survival-action movie with relentless enemies and nonstop tension

73/100

3) Mood & Genre (Optional)

Click to toggle genres and moods (multi-select).

Genres

ActionRomanceThrillerDramaComedyHorrorFamilyHistoricalCrimeSci-Fi

Moods

RevengeFeel-GoodTear-JerkerDanceEpicRoad TripSuspenseInspirational

Selected:

ActionCrimeRevenge

Results

10

Find My Movies

Clear

9. Vibe-Based Results - Top Matches

(Query - “Intense survival-action movie with relentless enemies and nonstop tension”)

Top results for vibe-driven search query and selected filters.

#1 BEST MATCH

GANGS OF WASSEYPUR - PART 2

हिंदी और वास्तविकता - १

Hindi

2012

2h 40m

★★★★☆7.8

(104 votes)

Crime

Thriller

Action

Director: Anurag Kashyap

Cast: Nawazuddin Siddiqui, Huma Qureshi, Richa Chadha

Match

87%

AI RESSAY

Why it matches: "Intense crime saga with relentless rivalries and nonstop tension. Directed by Anurag Kashyap, known for gritty narratives."

More Details

#2

GANGS OF WASSEYPUR - PART 1

हिंदी और वास्तविकता - १

Hindi

2012

2h 40m

★★★★☆7.1

(427 votes)

Action

Thriller

Crime

Director: Anurag Kashyap

Cast: Manoj Bajpayee, Richa Chadha, Tigmanshu Dhulia

Match

85%

AI RESSAY

Why it matches: Stunning performances and high-stakes action sequences.; "Part 1 sets the stage for the intense, relentless conflict. Anurag Kashyap's direction brings a raw, gritty atmosphere."

More Details

#3

AN ACTION HERO

हिंदी

2022

2h 8m

★★★★☆6.5

(75 votes)

Action

Thriller

Comedy

Director: Aniruddh Iyer

Cast: Ayushmann Khurrana, Jaideep Ahlawat, Gautam Joglekar

Match

86%

AI RESSAY

Why it matches: Complex characters and high tension make it a must-watch.; "Nonstop action with a relentless pace and intense survival. Akshay Kumar stars as a man on a mission."

More Details

#4

JAWAN

उर्दू

2023

2h 45m

Hindi

2023

2h 45m

★★★★☆7.1

(318 votes)

Action

Thriller

Director: Atlee

Cast: Shah Rukh Khan, Nayanthara, Vijay Sethupathi

Match

85%

AI RESSAY

Why it matches: Gripping narrative with high stakes and thrilling sequences.; "High-octane action with relentless enemies and intense drama. Starring Varun Dhawan, directed by Rohit Shetty."

More Details

10. Vibe-Based Results - Additional Matches

Continuation of vibe-query results with card-level details.

#5



DON 2

Hindi

2011

2h 13m

★★★★☆6.9

(260 votes)

Action

Thriller

Director: Farhan Akhtar

Cast: Shah Rukh Khan, Priyanka Chopra Jonas, Boman Irani

Match

94%

AI INSIGHT

Why it matches: Nonstop tension and spectacular stunts keep you on edge.; "Action-packed thriller with relentless enemies and high stakes. Starring Shah Rukh Khan, directed by Farhan Akhtar.

>

▼ More Details

#6



ANIMAL

Hindi

2023

3h 21m

★★★★☆6.1

(186 votes)

Action

Crime

Drama

Director: Sandeep Reddy Vanga

Cast: Ranbir Kapoor, Rashmika Mandanna, Anil Kapoor

Match

84%

AI INSIGHT

Why it matches: Intense plot twists and gripping revenge storyline.; "Intense survival story with relentless tension and nonstop action. Starring Ayushmann Khurrana, directed by Anand L. Rai.

>

▼ More Details

#7



SONCHIRIYA

Hindi

2019

2h 25m

★★★★☆8.0

(40 votes)

Crime

Action

Drama

Director: Abhishek Chaubey

Cast: Sushant Singh Rajput, Bhumi Pednekar, Ashutosh Rana

Match

83%

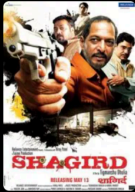
AI INSIGHT

Why it matches: Gripping narrative with high stakes and thrilling sequences.; "Gritty crime drama with relentless tension and intense action. Starring Sushant Singh Rajput, directed by Abhishek Chaubey.

>

▼ More Details

#8



SHAGIRD

Hindi

2011

2h 8m

★★★★☆6.6

(15 votes)

Drama

Action

Crime

Director: Tigmanshu Dhulia

Cast: Nana Patekar, Mohit Ahlawat, Rimi Sen

Match

82%

AI INSIGHT

Why it matches: High stakes and complex characters drive the narrative.; "Action-packed thriller with relentless tension and intense drama. Starring Emraan Hashmi, directed by Milan Luthria.

>

▼ More Details